



MATHEMATICS

GRADE TWO

INTRODUCTION

Mathematics is a subject that deals with numbers and symbols whose relationship is expressed in rules. Mathematics enables the learner to acquire competencies, values and positive attitudes which enable them to develop logical thinking, ability to apply the knowledge acquired, analyze situations and make rational decisions. The competencies that learners acquire enable them to continue with further education, lead a productive life and contribute to National economic development.

The subject is organized such that the content taught in one level becomes the prerequisite in the next level. The content therefore becomes more demanding as we move from one level to another.



VISION : an engaged, empowered and ethical citizen

MISSION: Nurturing every learner's potential

National Goals of Education

Education in Kenya should:

1. Foster nationalism, patriotism, and promote national unity

Kenya's people belong to different communities, races and religions and should be able to live and interact as one people. Education should enable the learner acquire a sense of nationhood and patriotism. It should also promote peace and harmonious co-existence.

2. Promote social, economic, technological and industrial needs for national development

Education should prepare the learner to play an effective and productive role in the nation.

a) Social Needs

Education should instill social and adaptive skills in the learner for effective participation in the family, community, national, regional and international development.

b) Economic Needs

Education should prepare a learner with requisite competences that support a modern and independent growing economy. This should translate into high standards of living for every individual.

c) Technological and Industrial Needs

Education should develop in the learner necessary competences for technological and industrial development for the nation in tandem with global trends.

3. Promote individual development and self-fulfillment



Education should provide opportunities for the learner to develop to the fullest potential. This includes development of one's interests, talents and character for positive contribution to the society.

4 Promote sound moral and religious values

Education should promote acquisition of national values as enshrined in the Kenya Constitution. It should be geared towards developing a self-disciplined and ethical citizen with sound moral and religious values.

5. Promote social equity and responsibility Education should promote social equity and responsibility. It should provide inclusive and equitable access to quality and differentiated education; including for learners with special educational needs and disabilities. Education should also provide the learner with opportunities to develop and practice shared responsibility and accountability through community service learning.

6. Promote respect for and development of Kenya's rich and varied cultures

Education should instill in the learner appreciation of Kenya's rich and diverse cultural heritage. The learner should value own and respect other people's culture as well as embrace positive cultural practices in a dynamic society.

7. Promote international consciousness and foster positive attitudes towards other nations

Kenya is part of the interdependent network of diverse peoples and nations. Education should empower the learner to respect, appreciate and participate in the opportunities within the international community. Education should also enable the learner to operate within the international community with full knowledge of the obligations, responsibilities, rights and benefits that this membership entails.

8. Promote positive attitudes towards good health and environmental protection



Education should inculcate in the learner the value of physical and psycho-social well-being for self and others. It should promote environmental preservation and conservation, including animal welfare, for sustainable development.



Learning Outcomes for Early Years Education

By the end of early years“ education, the learner should be able to:

1. Demonstrate basic literacy and numeracy skills for learning.
2. Communicate appropriately using verbal and/or non-verbal modes in a variety of contexts.
3. Demonstrate appropriate etiquette in social relationships.
4. Apply creativity and critical thinking skills in problem solving.
5. Explore the immediate environment for learning and enjoyment.
6. Practice hygiene, nutrition, sanitation, safety skills to promote health and wellbeing.
7. Demonstrate the acquisition of emotional, physical, spiritual, aesthetic and moral development for balanced living.
8. Demonstrate appreciation of the country's rich and diverse cultural heritage for harmonious co-existence.
9. Apply digital literacy skills for learning and enjoyment.

INTERRELATIONSHIP BETWEEN NATIONAL GOALS OF EDUCATION, LEVEL LEARNING OUTCOMES, SUBJECT GENERAL LEARNING OUTCOMES, SPECIFIC LEARNING OUTCOMES AND THE LEARNING EXPERIENCES

The level learning outcomes for early years education are derived from the National goals of education. It is from the early years education level learning outcomes that the General learning outcomes for mathematics activities are derived. The specific learning outcomes are then derived from the general learning outcomes from which the learning experiences are formulated.

The following table shows this relationship.

NATIONAL GOALS OF EDUCATION	LEVEL LEARNING OUTCOMES	SUBJECT GENERAL LEARNING OUTCOMES	SPECIFIC LEARNING OUTCOMES	SUGGESTED LEARNING EXPERIENCES
Promote sound moral and religious value	Demonstrate appropriate etiquette in social relationships	Apply measurement skills to find solutions to problems in a variety of context	Measure time using fixed units	Learners in pairs/groups to measure time taken to perform an activity using fixed units.
Promote individual development and self-fulfillment	Explore the immediate environment for learning and enjoyment	Demonstrate mastery of Number concepts by working out problems in day to	Appreciate number patterns as they skip on the number	Learners in pairs/groups to make number patterns and



		day life.	line.	share with other groups.
Forster nationalism, patriotism and promote national unity.	Communicate appropriately using verbal and/ or non-verbal modes in a variety of context	Describe properties of geometrical shapes and spatial relationship in real life experiences.	Identify rectangles, circles, triangles and squares.	Working in pairs/groups learners to identify and name the different shapes found in their classroom.

CORE COMPETENCIES

There are seven (7) core competencies to be achieved by every learner in basic education:

1. Communication and Collaboration
2. Critical Thinking and Problem Solving
3. Creativity and Imagination
4. Citizenship
5. Digital Literacy
6. Learning to Learn
7. Self-efficacy

Communication and collaboration

Communication is the act of transferring information from one place to another, whether vocally, visually, or non-verbally.

Collaboration: Collaboration is the process of two or more people or organizations working together to realize shared goals. Collaboration may require leadership, social within a decentralized or egalitarian group teams that work collaboratively in relation to gaining greater resources, recognition and motivation.

Self-efficacy

Self-efficacy is the person's belief about his or her capabilities to perform tasks or assignments that can change and transform his or her life. It determines how the person feels, thinks, behaves and motivates self positively. Self-efficacy has the potential to determine four major processes namely cognitive, motivational, and affective and selection processes.



Self-efficacy will enable learners to develop and nurture intra-personal skills and values such as self-awareness, self-esteem, confidence and personal integrity.

Critical thinking and problem solving

There are three types of thinking: reasoning, making judgments, and problem solving. When learners are empowered with critical thinking, they avoid being subjective, and use logic and evidence to arrive at conclusions. Critical thinking also facilitates exploring new ways of doing things and learner autonomy.

Critical thinking is important for lifelong learning. It helps learners to have an open mind and be ready to listen and appreciate information and opinions that may sometimes conflict with their earlier held beliefs and positions. Critical thinking and problem solving are useful for learners of all ages and in all subjects and disciplines offered in the basic education curriculum. For example, in the Sciences learners need to think critically about observations and patterns to develop ideas on how to solve problems. The competences are also important for solving problems in their lives and community, and ultimately achieve their potential which is the vision for basic education curriculum.

Creativity and Imagination

Imagination and Creativity refers to the ability to form new images and sensations in the mind, and to turn them into reality. It is the ability to imagine things that are not real, to form pictures in the mind, of things that one has not seen or experienced, and turning those pictures into real things. Imagination and Creativity on the other hand, is characterized by ability to perceive the world in new ways, to find hidden patterns, to make connections between seemingly unrelated phenomena, and to generate solutions.

In an educational set up, imagination and creativity refers to the ability of learners and their teachers to form images and ideas in their minds, and turn them into real, visible creations. Learners who are imaginative and creative are able to make life interesting for themselves and others around them. They are able to use the knowledge, skills and values acquired in the learning process to create new ideas which result in products that add value to their lives and to those of others around them.



Citizenship

Citizenship is the state of being vested with the rights, privileges, and duties of a citizen. It creates a sense of belonging and attachment to one's nation. A sense of citizenship helps to equip young people to deal with situations of conflict and controversy knowledgeably and tolerantly. They are able to understand the consequences of their actions, and those of the adults around them.

Digital literacy

Digital literacy is the knowledge, skills and behaviour in a broad use of digital content and devices. Such devices include mobile phones, smartphones, iPhone, tablets, laptops and desktops among others. All these are within the category of network devices. Digital literacy therefore focuses mainly on networking devices and should not be confused with computer literacy skills. However, traditional forms of literacy and computer literacy are enhancers in the acquisition of digital literacy skills.

Learning to learn

Learning to learn is the ability to pursue and persist in learning, to organize one's own learning, by effective management of time and information, both individually and in groups. This competence includes awareness of one's learning process and needs, identifying available opportunities, and the ability to overcome obstacles in order to learn successfully. This competence means gaining, processing and assimilating new knowledge and skill as well as seeking and making use of guidance. Learning to learn helps learners to build on prior learning and life experiences in order to use and apply knowledge and skills in a variety of contexts. There are four pillars of learning: Learning to know, learning to do, learning to be and learning to live together.



VALUES-BASED EDUCATION

Introduction

Values are standards that guide people on how to respond or behave in each situation. They influence how someone feels, acts and makes choices in life.

The responsibility for **nurturing values** rests with parents, school and the community. Everybody in the school has a role to play in helping the learner to nurture values. This approach is referred to as 'The whole school approach to value based education'.

The overall goal of values-based education is 'To nurture values in learners to become empowered, engaged and ethical citizens for positive and holistic transformation of society'. The teacher therefore has an important role to play in nurturing values through teaching.

Core Values

According to the competency based Curriculum there are eight core values to be nurtured in the learner namely love, responsibility, respect, unity, peace, patriotism, social justice, patriotism integrity **Refer to the appendix... Core Values.** The teacher is expected to identify value(s) to be nurtured in the learning experience in every strand. He/she is expected to constantly guide and engage the parents to reinforce the values learners have learnt in school at their home. Learners are expected to emulate exemplary behaviours and values from teachers, family members and the community at large hence the need for adults to exhibit good behaviour. Learners spend most of their formative years in school, which presents opportunities for the teacher to mould and reinforce values upon which the learner's character is formed.



PERTINENT AND CONTEMPORARY ISSUES

Introduction

Holistic, meaningful and learner centered education does not only focus on the subject content but also on the competencies the learner requires to effectively address the issues and challenges of everyday life thereby leading a fulfilled life and becoming a productive member of the society. Issues that have been identified as salient and therefore mainstreamed in the designs are referred to as Pertinent and contemporary issues (PCIs). Mainstreaming is the process of identifying suitable opportunity in the learning area or subject where PCI can be incorporated appropriately and taught alongside the subject matter. PCIs should be included in the schemes of work and in the lesson plans.

Classification of PCIs

PCIs have been classified into six broad areas comprising of several related salient issues as follows:

1. **Citizenship.** Peace Education, Integrity, ethnic and racial relations, social cohesion, patriotism, good governance, child's rights, child care and protection,
2. **Health Education Panel,** HIV and AIDS Education, Alcohol and drug abuse prevention, Life style diseases, personal hygiene, Common Communicable and non-communicable diseases and chronic, diseases
3. **Life Skills and Values Education:** Life Skills Education, Core Values and Human sexuality. See the appendices on Life Skills Education and Core values
4. **Education for Sustainable Development (ESD):** Environmental Education, Disaster Risk Reduction, Safety and security Education, Financial Literacy, Poverty eradication, Countering terrorism, extreme violence and radicalization, Gender issues in education and Animal Welfare Education
5. **Learner Support Programmes (Non Formal Education programmes):** Guidance services, Career guidance, counselling services, Peer education, mentorship, chaplaincy services, clubs and societies, Sports and games.
6. **Parental Empowerment and engagement and, Community Service learning:** Parental empowerment and engagement and, Community and Service learning



During lesson preparation, the teacher has to be deliberate in identifying specific PCIs and planning how they shall be incorporated in the lesson. He /She should ensure activities on PCIs are undertaken and covered during the learning experiences.

The teacher is expected to constantly guide and engage the parents to reinforce what the learner has learnt in school at home. The school community and the larger community should provide conducive environment for learners to apply competencies they have learnt on PCIs to address issues at home and in the society. Please refer to the sample lesson plan given in the handbook/Teachers guide.

PCIs are supposed to not only be integrated in the lesson but in other programmes and activities in the school, home and in the community. Acquisition of knowledge, values and skills in various PCIs will enable the learner to translate what they have learnt to real life situation both in and outside school.

Differentiated Learning

Every learner learns differently. Technically, individual learners have preferential way in which they absorb processes, comprehend and retain information. It is therefore important for educators to understand the differences in their learners' preferential way of learning, so that they can implement best practice strategies into their daily activities, curriculum and assessments.

Differentiated Learning is effective instruction that is responsive to learners' readiness, interests and learning preferences. It enables learners to build new learning through connections to existing knowledge and preferred ways of learning. Readiness is the learner's starting point for learning, relative to the concept being studied while his or her interests enhance the relevancy of learning by linking new information to his or her experience and enthusiasm. The learning preferences refer to different ways in which learners prefer to acquire process and work with information.



Learning preferences are influenced by gender, culture, the classroom environment, learning styles and multiple intelligences. Teachers are required to use a range of instructional and assessment strategies to meet the needs of different learners. Learners are assessed before, during and after learning. Assessments inform next steps for both teacher and learner. All differentiated instruction activities are equally engaging and respectful and take approximately the same amount of time.

Differentiated learning allows for flexibility in creating and altering instructional plans in response to learners. It involves expanding instructional routines and skills where learning experiences are introduced to learners in different ways of learning. Differentiated learning provides effective teaching that involves providing different learners with different avenues to learning.

USE OF INFORMATION COMMUNICATION TECHNOLOGY (ICT)

ICT entails the use of technology in information delivery. Consequently ICT in education entails teaching and learning using technology devices and material.

The resources used in ICT are mainly digital, electronic and Portable Document Format (PDF) files. The bulk of ICT education materials in KICD are radio, television, e-learning programmes and PDFs.

- i) **Radio programmes** - involve audio recordings disseminated through radio/electronic devices
- ii) **Television (Video) programmes** - audio-visual content that is mainly video based and disseminated through television and other electronic devices.
- iii) **E – learning (computer based) programmes** - electronic learning materials disseminated through computer /smart T.Vs technology.
- iv) **PDFs** - electronically converted images from print material.

The education ICT content is stored in various electronic formats including offline modes such as CDs, DVDs, Flash disks, Memory cards and online formats.

The KICD electronic content can be accessed from the following Digital Literacy Programme devices in the schools:

- Teacher Digital Device (TDD)



- Learner Digital Device (LDD)
- Digital Content Server & Wireless Router (DCSWR)
- External Hard Disk

When chosen and used appropriately, the electronic material will enable the learner use modern technologies thus making ICT technology a delivery tool for all learning areas. This added advantage to teaching and learning enhances learning by:-

- i. Supplementing classroom instruction
- ii. Stimulating and motivating the learner
- iii. Enhancing concepts acquisition
- iv. Arousing learners interest and promoting active participation during the lesson
- v. Saving time used to explain concepts
- vi. Enhancing skills development
- vii. Enabling the teacher to cater for individual differences
- viii. Reaching learners by multisensory presentation

Research findings indicate that retention of knowledge and skills is highly accomplished through the three sensory domains of audio, audio-visual and tactile. The teachers are therefore encouraged to use quality Educational Resources. The endeavour will enable teachers to access resources that will address the needs of the learner in order to make learning interesting and more meaningful. When choosing the appropriate resource, the teacher should take note of the following:-

- a) Relevance to the content being taught
- b) Accuracy
- c) Stimulation of learner's imagination and enhancement of concepts acquisition
- d) Appropriateness to the level of the learner
- e) Capability of capturing the learner's interest for active participation
- f) Durability and safety for learners use
- g) Application of new knowledge and skills
- h) Conformity to set standards
- i) Awareness creation to teachers on the latest pedagogical practices
- j) Interactivity and sustainability



The use of technology to teach and learn is a major educational reform.

It can be used as a tool for learning. It is a valuable resource which can be used:

- ✓ As a method for teaching
- ✓ For assessment
- ✓ To introduce the lesson and
- ✓ To enhance what was learnt

Information and Communication Technologies play an increasingly important role in the way we communicate, learn and live. The challenge is to effectively harness these technologies in a way that serves the interests of learners in their learning environment.

ICT should contribute to access to education, equity in education and the delivery of quality learning and teaching. It is not meant to replace the teacher. Instead, consideration must be given to the subject matter, the learning objectives and outcomes, the characteristics of the learners, and the learning context in order to arrive at the optimum mix of instructional and delivery methods. Moreover, different technologies are typically used in combination rather than as the sole delivery mechanism.

PARENTAL EMPOWERMENT AND ENGAGEMENT

Children start learning before they go to school. Parents, other family members and care-givers are a child's first and most important educator. They can be a positive influence and help child do well at school. Parents therefore require empowerment and opportunity to get involved throughout their children's schooling.

As soon as children enrol for formal learning, teachers have a responsibility to create awareness and provide an enabling environment for parents to engage in their child's learning. Involving parents can have very positive impact on education outcomes. The following are suggested ways teachers can employ to empower and engage parents.

Parents can be sensitized on:

1. Parental roles and responsibilities with regard to education according to the constitution and other national and school policies.



2. The need to ensure the child is given proper care and protection to experience normal patterns of growth and development.
3. The importance of providing basic care, nutrition, clothing, shelter and access to good health.
4. Importance of early identification of disability and prompt referral for early intervention to mitigate the disability in order to avoid further delay or distorted development.
5. The need to have a safe and conducive environment to enable a child to be creative and innovative.
6. Importance of creating a family culture where values taught in school are reinforced at home.
7. The need to equip children with basic self-help skills e.g. toilet training, eating and table manners/etiquette and dressing up.
8. The fact that children need to develop fine motor skills so they can do activities such as scribbling, painting, clothing and opening buttons.
9. The importance of providing resources e.g. school uniform, toys and other items required by the teacher in various learning areas and subjects.
10. The importance of carrying out roles and tasks assigned by the teacher for instance reinforcing a skill, value or provision of extra information on an issue.
11. Reading and narrating to children stories.
12. Importance of seeking information from the school head and teachers about the formal, informal and the non-formal aspects of the child's potential at school.

Parents play a very important role in determining the success of a child's education. Teachers in liaison with the head teacher should from time to time have fora to discuss strategies they can employ to empower and engage parents in their school activities.

Professional documents and their use – (TT)

In order to plan for effective delivery of the curriculum, it is crucial that teachers plan their work well. Professional documents are used to organise curriculum implementation. Kenya Institute of Curriculum Development develops **curriculum designs**. These are curriculum document which



usually contains specific information about the course such as, details on the expected learning outcomes, what is to be taught, how it can be taught, and suggestions for assessment, resources and time allocation. It is therefore a vital document that the teacher must use in the teaching and learning process. The teacher should use the curriculum designs while preparing schemes of work and lesson plans.

Schemes of Work are developed by the implementing teacher to organise how the curriculum will be implemented in a given year. It is developed from the curriculum designs. It helps the teacher distribute what is to be covered within the available time and how it shall be done. It therefore shows how the planned curriculum shall be distributed or organised within the time allocated. The teacher who is expected to teach the activity area should develop the scheme of work as he/she can easily contextualise what is feasible for implementation in his/her context. He or she is able to tell what resources are available for him or her to use in the learning process. A sample Scheme of Work (SoW) is appended at the back of this book.

Lesson plans are developed from the schemes of work. They describe the actual teaching and learning experiences that the teacher will facilitate during the lesson. It provides clear instructions on how the class will be organised, the resources that will be used, and the activities that will take place within the time allocated. It also shows how values, PCIs and competencies shall be mainstreamed in the lesson. Lesson plans are usually propositions made by the teacher on how the lesson shall run, however, the way in which the lesson may run could change spontaneously during the lesson. Lesson plans should be developed by the teacher who will facilitate the lesson as they are better placed to know the type of learners they have, the resources available to them and the context in which the learning shall take place. A sample lesson plan for a competency based curriculum is provided in the appendix of this book.

After facilitating a lesson the teacher should indicate what was covered in the lesson and how well the learners achieved the learning outcomes envisaged. This document is called the **record of work book**. The record of work book is important as it indicates what the learners have achieved within a given time. It also enables the school management to track what is happening in the classes in their schools. In case the teacher is absent or leaves the school, the record of



work book enables the replacing teacher to pick up from where they left and effectively continue implementing the curriculum. Record of work books should be signed by the teachers and also regularly signed by the school management.

In order to track learner's performance the teacher should also have an **assessment record book**. The book will track the achievement of learning outcomes and competencies acquired by the learners. It will guide the teacher on which learner needs special attention to acquire the skills and competencies stated in the curriculum designs. The information will also be used when reporting learner's progress to their parents. A sample of an assessment report is also provided in the appendix of this book.

HOW TO USE THIS HANDBOOK

This handbook contains the following:

- Strands
- Sub strands
- Specific learning outcomes
- Suggested learning experiences
- Issues in the reformed curriculum
- Assessment

STRANDS AND SUB- STRANDS

This handbook has discussed the concepts in first term of Early Grade Mathematics (EGM) under the PRIEDE project. The handbook guides the teacher on how to integrate issues in the reformed curriculum with those in EGM.



STRAND	SUB- STRAND
NUMBERS	WHOLE NUMBERS
	ADDITION
	SUBTRACTION
MEASUREMENT	MASS
	TIME
GEOMETRY	SHAPES

In the suggested learning experiences the following issues have been discussed:

- Core competencies
- Pertinent and contemporary issues
- Values
- Links to other subjects
- Suggested non-formal activities to support learning
- Suggested community service learning
- Suggested assessment

The teacher should refer to the key inquiry questions stated under each sub strand in the curriculum design during the teaching and learning process. On assessment the teacher should always refer to the assessment rubrics stated in the designs as well as the suggested methods indicated in the designs under suggested assessment.

A report card template has been included. This template will assist the teacher in reporting the learner's performance both in relation to the achievement of content in the classroom as well as the achievement of the core competencies.

SECTION TWO

ESSENCE STATEMENT

Numeracy is a foundational skill that prepares the learner for number work, mathematics in higher levels of schooling and mathematical approaches in all aspects of life. Numeracy activities involve identification and value placement of mathematical numerals, basic mathematical operations as well as measuring and describing shapes.



SUBJECT LEARNING OUTCOMES:

1. Demonstrate mastery of Number concepts by working out problems in day to day life.
2. Apply measurement skills to find solutions to problems in a variety of contexts
3. Describe properties of Geometrical shapes and spatial relationships in real life experiences

ISSUES IN THE REFORMED MATHEMATICS CURRICULUM INTEGRATED WITH EARLY GRADE MATHEMATICS (EGM)

Introduction

When teaching various concepts in Early Grade Mathematics, it is important to emphasize the following issues in the reformed curriculum:

- i) Core competencies to be developed
- ii) Links to pertinent and contemporary issues (PCI's)
- iii) Links to values
- iv) Links to other subjects
- v) Suggested non formal activity to support learning
- vi) Suggested community service learning activities, and
- vii) Suggested assessment

When teaching concepts in EGM it is important to refer to the information given in this handbook for guidance on how the above issues can be integrated.

1.0 STRAND: Numbers

1.1 Sub Strand: whole numbers

Specific Learning Outcomes

- a) Count numbers forward and backward up to 100
- b) Identify place value of numbers up to 100
- c) Read numbers 1-100 in symbols
- d) Read and write numbers 1-20 in words



- e) Work out missing numbers in number patterns up to 100
- f) Appreciate number patterns as they skip on the number line

Suggested learning Experiences

In whole numbers we have the following learning experiences:

- a) Learners in pairs/groups to count in 2's and 5's forward and backward starting from any point
- b) Learners in pairs/groups to count their fingers and toes in 2's and 10's forward and backward starting at any point.
- c) Learners in pairs / groups to discuss place value of digits in given numbers up to 100
- d) Learners in pairs to read numbers 1-100 in symbols
- e) Learners to read and write numbers 1-20 in words
- f) Learners to play digital games on whole numbers
- g) Learners to work out missing numbers in patterns up to 100
- h) Learners in pairs/groups to make number patterns and share with other groups

When learners are counting, reading and writing numbers forward and backward in groups, they will enhance their imagination and creativity as well as develop communication and collaboration competencies. In identifying place value of digits, comparing and ordering numbers using place value they develop the desire to learn and think critically and build self-efficacy. Learners could collect concrete objects, count and group in tens in the development of place value concept. The teacher should guide the learners on the safety of the objects being used. While working in groups, it is expected that the learners will be responsible and respect each other in order to achieve their goal.

The concept of whole numbers is applicable in other learning areas namely language, environmental, movement and creative activities. Learners may assist in arranging furniture in rows and columns at school and community functions using patterns. Assessment may be oral, written or by observation following the assessment rubrics in the curriculum design.



1.0 Strand: Numbers

1.2 Sub Strand: Addition

Specific Learning Outcomes

- a) Add a 2- digit number to a 1- digit number without and with regrouping with sum not exceeding 100
- b) Add 3-single digit numbers up to a sum of 20
- c) Add a 2-digit number to a 2-digit number without and with regrouping, with sums not exceeding 100
- d) Work out missing numbers in patterns involving addition of whole numbers up to 100

Suggested Learning Experiences

Under addition we have the following learning experiences:

- a) Learners in pairs to write addition sentences given in horizontal form vertically according to place value
- b) Learners to add a 2- digit number to a 1- digit number without and with regrouping
- c) Learners to practice addition by skipping on the number line
- d) Learners in pairs/groups to collect different and safe objects and use them in addition of 3-single digit numbers
- e) Learners in pairs/groups to practice breaking numbers apart to make a 10.
- f) Learners in pairs to come up with different ways of adding two 2-digit numbers without and with regrouping
- g) Learners to play digital games involving addition
- h) Learners in groups to make patterns using numbers up to 100

When working out addition, the following core competencies could be developed: learning to learn, communication and collaboration, critical thinking and problem solving. Learners at this level are expected to work out problems involving addition of up to 2-digit numbers without regrouping in different contexts. As learners use concrete objects in working out addition, it is important for the teacher to emphasize safety of the materials being used. While working in groups it is expected that the learner will acquire a sense of respect, responsibility and unity of purpose. The concept of addition is applicable in other learning areas such as hygiene and nutrition, environmental and language activities. Learners could be involved in cultural dances



where different formations of number patterns are used, both at school and home. Learners could also assist in getting the total number of items in their homes, children's homes as well as in community functions. Assessment may be oral, written or observation and should be guided by the assessment rubrics as stated in the curriculum designs.

1.0 Strand: Numbers

1.3 Sub strand: Subtraction

Specific Learning Outcomes

- a) Subtract up to 2- digit numbers without regrouping
- b) Use the relationship between addition and subtraction in working out problems
- c) work out missing numbers in subtraction of up to 2- digit numbers
- d) Work out missing numbers in patterns involving subtraction up to 100

Suggested Learning Experiences

Under the sub strand on subtraction we have the following learning experiences:

- a) Learners in pairs /groups to subtract single digit numbers by comparing groups of objects
- b) Learners to subtract up to 2-digit numbers without regrouping in horizontal and vertical forms
- c) Learners to discuss the relationship between addition and subtraction using number families.
- d) Learners to work out missing numbers in subtraction of up to 2- digit numbers
- e) Learners to play digital games involving subtraction
- f) Learners to work out missing numbers in number patterns involving subtraction

In subtraction, learners will be involved in solving problems in different context as individuals and in groups. As they work in groups, the learners will enhance their interpersonal relationship, effective communication, social cohesion and friendship formation skills. In individual activities they may acquire competencies like self-efficacy, learning to learn, critical thinking and problem solving. The group work activities will also lead to acquisition and nurturing of such values like respect, responsibility and unity. The concept of subtraction is also applicable in language and environmental activities. Learners may participate in cleanup activities in school or in the community where they could observe reduction of waste. The teacher should ensure learners



safety as they undertake such activities. Assessment may be oral, written or observation and should be guided by the assessment rubrics as stated in the curriculum designs.

2.0 Strand: Measurement

2.1 Sub Strand: Mass

Specific Learning Outcomes

- a) compare mass of objects directly,
- b) conserve mass through manipulation,
- c) measure mass using arbitrary units

Suggested Learning Experiences

Under Mass, we have the following learning experiences:

- a) Learners in pairs/groups use safe objects to identify those heavier than, lighter than or same.
- b) Learners to use two objects of equal mass and a beam balance to demonstrate that change of shape does not change the mass of an object.
- c) Learners in pairs/groups to use an identified mass to compare the mass of other objects using the words heavier than, lighter than or same as.

Mass should be taught at this level by comparing objects in terms of ‘heavier than’, ‘lighter than’ or ‘same as’. This could be done by lifting objects or using a simple beam balance. As learners work in pairs/groups to identify objects of different masses, (‘lighter than’, ‘same as’ or ‘heavier than’), they develop competencies such as communication and collaboration, critical thinking and problem solving, imagination and creativity and digital literacy.

The teacher should guide the learners on safe objects during the process of comparison and also instill values such as honesty and responsibility in their observations. The teacher should advise them to wash their hands with soap and clean water after the lesson.

The concept of mass is relevant in environmental, language, movement and activities. For example in environmental activities the learners may be involved in carrying various objects from one place to another and be able to distinguish the heavier, lighter or same as items while carrying. As a non-formal learning activity the learners may assist in carrying light objects such as water, grains at home and books in the school store. A community service learning activity suggested for this learning outcome may include assisting an aged neighbor in carrying light masses of animal feeds. The teacher may assess learners through oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.



(Disclaimer: This content is found in grade 1 of the curriculum design. However, it's still possible to address it in Grade 2 EGM.)

2.0 Strand: Measurement

2.1 Sub Strand: Time

Specific Learning Outcomes

- a) Relate the months of the year with various activities
- b) Recite the number of days in each month of the year
- c) Measure time using arbitrary units
- d) Measure time using fixed units
- e) Identify the clock face
- f) Read, tell and write time by the hour

Suggested Learning Experiences

Under Time we have the following learning experiences:

- a) Learners in pairs/groups to discuss activities that take place in the months of the year.
- b) Learners in pairs/groups to sing songs, rhymes related to number of days in the months of the year.
- c) Learners in pairs/groups to measure time taken to perform an activity using arbitrary units.
- d) Learners in pairs/groups to measure time taken to perform an activity using fixed units.
- e) Learners to discuss places where they have seen clocks displayed as well as how they look like.
- f) Learners to observe a clock face and discuss the minute hand and the hour hand.
- g) Learners to discuss how to read, tell and write time by the hour using both the analogue and digital clock

The concept of time as a means of measurement is introduced to learners through the analogue clock face. At this level learners are tasked to tell the difference between the hour and minute hand and then be able to read, tell and write time by the hour. This involves group activities that lead to promotion of team work, good communication and collaboration skills. Imagination and creativity, self- efficacy, digital literacy are enhanced as learners perceive the world and make connections of phenomena by assimilating knowledge and functions of the minute and hour-hand of the clock face to daily living. When they successfully internalize the approximate length of an hour then they will get assertive to effectively plan and manage time and activities both at school and home.



As part of non-formal learning they may help the school bell ringer by observing time for school activities, and the informal learning by reading time on as many clock faces that they come into contact with as possible e.g. city clock, clocks in social places, hospitals and even at home. As a community service learners would help train people/family on how to read and use the analogue clock.

Time is a concept found in other learning areas such as religious and language activities. Assessment may be oral, written or by observation using the assessment rubrics in the curriculum design.

3.0 Strand: Geometry

3.1 Sub Strand: Shapes

Specific Learning Outcomes

- a) Identify rectangles, circles, triangles, ovals and squares,
- b) Appreciate making patterns involving rectangles, circles, triangles, ovals and squares.

Suggested Learning Experiences

Under shapes we have the following learning experiences:

- a) Learners in pairs/groups to sort and group items of different shapes.
- b) Learners in pairs/groups to discuss types of lines making different shapes.
- c) Learners to identify and name the different shapes found in their classroom.
- d) Learners to make patterns of their choice using the five shapes.
- e) Learners in groups to make patterns, colour them and share with other groups.
- f) Learners to make patterns using digital devices.

The process of exposing learners to recognition and identification of shapes (circles, rectangles, squares and triangles) involves group activities through which learner's interpersonal relationships are enhanced and thereby improving their communication and collaboration skills. This may extend to the formation of friendship and development of social cohesion that help learners to learn to support each other and share during digital learning.



The cohesion may also lead to discovery of each other's talents which may positively translate into meaningful Clubs and Societies that help promote education for sustainable development. Clubs and Societies also help to foster respect, integrity, patriotism and social justice in the community hence reducing radicalization, violence and terrorism.

Making patterns involving the geometrical shapes and alphabetical letters helps learners in developing critical thinking and problem solving skills, imagination and creativity skills as they draw, cut out and arrange the shapes.

Teacher is advised to guide and caution learners on safety of handling cutting materials. On the other hand the learner is expected to observe responsibility in engaging and fulfilling assigned roles and duties in school, home and community. For further learning, they can identify and appreciate circular, rectangular, square and triangular shapes in school, home and community. They too can make cut-outs of different shapes and use them for labelling e.g. price tags or showing direction e.g. school library.

The concept of shapes is also applicable in other learning areas namely, language, movement and creative activities. Assessment may be oral, written or by observation using the assessment rubrics in the curriculum design.

TERM TWO

STRANDS AND SUB- STRANDS

This handbook has discussed the concepts in second term of Early Grade Mathematics (EGM) under the PRIEDE project. The handbook guides the teacher on how to integrate issues in the reformed curriculum with those in EGM.

STRAND	SUB- STRAND
NUMBERS	WHOLE NUMBERS
	ADDITION
	SUBTRACTION
	MULTIPLICATION
MEASUREMENTS	LENGTH
	CAPACITY



	MONEY
GEOMETRY	SHAPES

In the suggested learning experiences the following issues have been discussed:

- Core competencies
- Pertinent and contemporary issues
- Values
- Links to other subjects
- Suggested non-formal activities to support learning
- Suggested community service learning
- Suggested assessment

The teacher should refer to the key inquiry questions stated under each sub strand in the curriculum design during the teaching and learning process. On assessment the teacher should always refer to the assessment rubrics stated in the designs as well as the suggested methods indicated in the designs under suggested assessment.

A report card template has been included. This template will assist the teacher in reporting the learner's performance both in relation to the achievement of content in the classroom as well as the achievement of the core competencies.

SECTION TWO

ESSENCE STATEMENT

Numeracy is a foundational skill that prepares the learner for number work, mathematics in higher levels of schooling and mathematical approaches in all aspects of life. Numeracy activities involve identification and value placement of mathematical numerals, basic mathematical operations as well as measuring and describing shapes.

SUBJECT LEARNING OUTCOMES:

- 1) Demonstrate mastery of Number concepts by working out problems in day to day life.
- 2) Apply measurement skills to find solutions to problems in a variety of contexts



- 3) Describe properties of Geometrical shapes and spatial relationships in real life experiences

ISSUES IN THE REFORMED MATHEMATICS CURRICULUM INTEGRATED WITH EARLY GRADE MATHEMATICS (EGM)

Introduction

When teaching various concepts in Early Grade Mathematics, it is important to emphasize the following issues in the reformed curriculum:

- i) Core competencies to be developed
- ii) Links to pertinent and contemporary issues (PCI's)
- iii) Links to values
- iv) Links to other subjects
- v) Suggested non formal activity to support learning
- vi) Suggested community service learning activities, and
- vii) Suggested assessment

When teaching concepts in EGM it is important to refer to the information given in this handbook for guidance on how the above issues can be integrated.

1.0 Strand: Numbers

1.1 Sub-strand: whole numbers

Specific Learning Outcomes

- a) Count numbers forward and backward up to 100
- b) Identify place value of numbers up to 100
- c) Read numbers 1-100 in symbols
- d) Read and write numbers 1-20 in words
- e) Work out missing numbers in number patterns up to 100
- f) Appreciate number patterns as they skip on the number line

Suggested learning Experiences

In whole numbers we have the following learning experiences:



- a) Learners in pairs/groups to count in 2's and 5's forward and backward starting from any point
- b) Learners in pairs/groups to count their fingers and toes in 2's and 10's forward and backward starting at any point.
- c) Learners in pairs / groups to discuss place value of digits in given numbers up to 100
- d) Learners in pairs to read numbers 1-100 in symbols
- e) Learners to read and write numbers 1-20 in words
- f) Learners to play digital games on whole numbers
- g) Learners to work out missing numbers in patterns up to 100
- h) Learners in pairs/groups to make number patterns and share with other groups

When learners are counting forward and backwards, representing numbers in tally form, reading and writing numbers 1- 100 in symbols and comparing numbers, they develop the competencies of creativity and imagination by identifying hidden patterns as well as exploring new ways of doing things. While working in groups learners develop communication and collaboration skills. Values such as responsibility, love and respect are fostered as learners work in groups.

The concept of whole numbers is applicable in other subject areas namely language, environmental, movement and creative activities. Using the knowledge of tallying the learners could assist in counting and arranging the number of utensils at home and books in the school store. Assessment may be oral, written or observation following the assessment rubrics in the curriculum design.

1.0 Strands: Numbers

1.2 Sub Strand: Addition

Specific Learning Outcomes

- a) Add a 2- digit number to a 1- digit number without and with regrouping with sum not exceeding 100
- b) Add 3-single digit numbers up to a sum of 20
- c) Add a 2-digit number to a 2-digit number without and with regrouping, with sums not exceeding 100
- d) Work out missing numbers in patterns involving addition of whole numbers up to 100

Suggested Learning Experiences



Under addition we have the following learning experiences:

- a) Learners in pairs to write addition sentences given in horizontal form vertically according to place value
- b) Learners to add a 2- digit number to a 1- digit number without and with regrouping
- c) Learners to practice addition by skipping on the number line
- d) Learners in pairs/groups to collect different and safe objects and use them in addition of 3-single digit numbers
- e) Learners in pairs/groups to practice breaking numbers apart to make a 10.
- f) Learners in pairs to come up with different ways of adding two 2-digit numbers without and with regrouping
- g) Learners to play digital games involving addition
- h) Learners in groups to make patterns using numbers up to 100

In addition of up to 2-digit numbers with regrouping, the learner is empowered to think, reason and make judgments in problem solving. While working out additions involving zero, the learner is required to make connections between seemingly unrelated phenomena to generate solution, hence enhancing creativity and imagination. Learners could use concrete objects as well as digital games in working out addition. It is important for the teacher to emphasize safety of the materials used by the learners.

As learners are required to work in groups, they will develop unity and respect for each other which will foster peace and love among themselves. The concept of addition is applicable in other learning areas such as hygiene and nutrition, language as well as movement and creative activities. Learners may be involved in planting flowers in patterns both at school and home. Learners could be guided in getting the total enrolment of the school per class in terms of boys and girls or the total number of male and female attending a community function. Assessment may be oral, written or observation and should be guided by the assessment rubrics as stated in the curriculum designs.

1.0 Strand: Numbers

1.3 Sub strand: Subtraction

Specific Learning Outcomes



- a) Subtract up to 2- digit numbers without regrouping
- b) Use the relationship between addition and subtraction in working out problems
- c) work out missing numbers in subtraction of up to 2- digit numbers
- d) Work out missing numbers in patterns involving subtraction up to 100

Suggested Learning Experiences

Under the sub strand on subtraction we have the following learning experiences:

- a) Learners in pairs /groups to subtract single digit numbers by comparing groups of objects
- b) Learners to subtract up to 2-digit numbers without regrouping in horizontal and vertical forms
- c) Learners to discuss the relationship between addition and subtraction using number families.
- d) Learners to work out missing numbers in subtraction of up to 2- digit numbers
- e) Learners to play digital games involving subtraction
- f) Learners to work out missing numbers in number patterns involving subtraction

In this sub strand learners are exposed to various concepts namely subtraction using number families, solving word questions involving subtraction and also completing number patterns that involve subtraction. During this process learners think critically about information given (word problems), processes involved in the word problem and logical patterns therein. This leads to the development of problem solving skills as well as the ability to analyze given patterns in order to determine the next number in the sequence. As Learners perform tasks, they develop intra and inter-personal skills leading to improvement of self esteem, confidence and integrity.

The learners could assist parents and youth agricultural based clubs in planting seeds in seed beds in rows and columns. Subtraction is a concept found in other learning areas such as language, environmental, creative and movement activities. Assessment in this area may include oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.

1.0 Strand: Numbers



1.4 Sub strand: Multiplication

Specific Learning Outcomes

- a) Represent multiplication as repeated addition using numbers 1, 2, 3, 4 and 5 up to five times
- b) Write repeated addition sentences as multiplication using 'x' sign
- c) Multiply single digit numbers by 1, 2, 3, 4, 5 and 10

Suggested Learning Experiences

Under multiplication we have the following learning experiences:

- a) Learners in pairs/groups to use counters to represent multiplication as repeated addition
- b) Learners in pairs/groups to use number lines to represent multiplication as repeated addition
- c) Learners to use 'x' sign in writing repeated addition sentences as multiplication
- d) Learners to multiply single digit numbers by 1, 2, 3, 4, 5 and 10
- e) Learners to play digital games involving multiplication
- f) Learners could visit the local market to see how fruits are arranged in groups of 3's, 4's, 5's or 10's a certain number of times

The concept of multiplication at this level is introduced as repeated addition using objects and also as repeated addition using numbers. The multiplication sign and multiplication sentences are also introduced. Later learners are exposed to multiplication of ten by single digit numbers which is equivalent to counting in tens using objects.

This involves group activities where learners improve on communication and collaboration skills as they work together towards a common goal. The core values enhanced are unity, as they work harmoniously; respect through idea of open mindedness and willingness to understand and appreciate each other's thinking which later leads to integrity and social cohesion.

In school learners can help arrange chairs and desks according to rows and columns and help getting total number of learners present. At home or community they could help getting total number of people attending a given function or religious activity.

Multiplication concept is found in other learning areas such as environmental, religious and movement and creative activities. The teacher may assess learners through oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.



2.0 Strand: Measurement

2.1 Sub strand: Length

Specific Learning Outcomes

- a) Measure length using fixed units
- b) Identify the metre as a unit of measuring length
- c) Measure length in metres

Suggested learning experiences

In this sub strand we have the following learning experiences:

- a) Learners in pairs/groups to use sticks of equal length to measure different lengths, record and discuss the results
- b) Learners in pairs/groups to measure length using stick of different lengths, including 1-metre sticks and identify the 1-metre sticks
- c) Learners to make 1-metre stick and use them in measuring various lengths within the classroom, record and discuss the results
- d) Learners to play digital games involving length in metres.

The learner at this level is required to use a 1-metre stick as the unit of measuring length.

Learners in groups/pairs should make 1-metre sticks and use them to measure and record different lengths within the classroom. Learners could play digital games involving lengths in metres. During these activities the learner will develop teamwork, realise capabilities to perform tasks hence develop and nurture intra-personal skills as well as critical thinking and problem solving competencies. Group work enhances unity, responsibility and integrity as they measure and record different items in the classroom. The teacher should guide learners on the safety of the sticks used and care when using them. The concept of length is applicable in other learning areas such as language, environmental as well as movement and creative activities. The learners could assist in marking small playfields for games and uniform spacing of seedlings during community tree-planting activities.

The teacher may assess learners through oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.



2.2 Strand: Measurement

2.2 Sub Strand: Capacity

Specific Learning Outcomes

- a) compare capacity of containers directly,
- b) conserve capacity through manipulation,
- c) measure capacity using arbitrary units

Suggested Learning Experiences

Under mass we have the following learning experiences:

- a) Learners to empty and fill water in different containers to establish which holds more, which holds less and which holds the same.
- b) Learners to identify and compare containers which holds more, less or same as.
- c) Learners to fill containers of different shapes and sizes with water then empty into others so as to establish that some containers can hold the same amount although their shapes are different.
- d) Learners to be given water, same size basins and different small containers. The learners to count the number of small containers they use to fill the basin.

The concept of capacity as a means of measurement at this level focuses mainly on measuring capacity using arbitrary units and then using fixed unit to measure capacity.

As learners work in groups carrying out activities of filling smaller containers and emptying into bigger ones, they develop communication and collaboration, critical thinking and problem solving, creativity and imagination competencies. Through discussions, predictions and verification of their predictions, learners would be motivated to learn more. Values such as unity of purpose regardless of their divergent views, integrity as they embrace the virtue of honesty and social justice in measurement could be achieved. The teacher should guide learners on safety during the collection of materials and activities thereof. At school learners could use containers to water flowers or seedbeds and draw water for cleaning classrooms. At home and community, learners could use same or similar containers to draw water for the family, their animals and for those advanced in age.



The concept of capacity may be linked to other learning areas such as environmental and religious activities. The teacher is advised to assess learners through oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.

2.0 Strand: Measurement

2.4 Sub Strand: Money

Specific Learning Outcomes

- a) identify Kenyan currency coins and notes up to sh.100,
- b) count money in sh.1, sh.5, sh.10, sh.20, sh.40, sh.50 up to sh.100,
- c) represent same amount of money in different denominations,
- d) relate money to goods and services up to sh.100,
- e) differentiate between needs and wants in real life context,
- f) appreciate spending and saving of money in real life situations.

Suggested Learning Experiences

Under money we have the following learning experiences:

- a) Learners in pairs/groups to sort out Kenyan currency coins and notes according to their features up to sh.100.
- b) Learners in groups to put different coins and notes together and separate them according to their values and features.
- c) Learners in pairs/groups to count money in sh.1, sh.5, sh.10, sh.20, sh.40, sh.50 up to sh.100.
- d) Learners in pairs/groups to make same amount of money using different denominations.
- e) Learners in pairs/groups to discuss items they cannot do without and those that are necessary but they can do without up to a value of sh.100.
- f) Learners in pairs/groups to classify needs and wants.
- g) Learners to discuss the importance of saving.
- h) Learners to play digital games involving money.
- i) Learners could record a video during a role play of classroom shopping activities for replay and discussion later

While learning about money, learners will be required to identify Kenyan currency notes up to sh.500 in terms of size, value and features using real money and pictures, work out problems involving addition and subtraction of shillings and cents without regrouping. Learners appreciate their country while identifying the features in the currency hence fostering patriotism. This also brings a sense of belonging and attachment to one's country among the learners as well portrays a sense of responsibility and integrity. Learners will be required to distinguish between needs and wants in real life contexts. They will therefore be expected to make judgement.



Knowledge on money is applicable in many learning areas such as language and environmental activities. Learners can be called upon to assist in counting money collected for a class project. Learners may also be involved in counting money offered in religious functions as an activity for community service learning. They could also establish a class savings account or home bank and keep track of the money collected for a worthy cause.

The teacher may assess learners through oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.

3.0 Strand Geometry

3.1 Sub-Strand Shapes

Specific Learning Outcomes

- a) identify rectangles, circles, triangles, ovals and squares,
- b) appreciate making patterns involving rectangles, circles, triangles, ovals and squares

Suggested Learning Experiences

Under this sub strand on shapes we have the following learning experiences:

- a) Learners in pairs/groups to sort and group items of different shapes.
- b) Learners in pairs/groups to discuss types of lines making different shapes.
- c) Learners to identify and name the different shapes found in their classroom.
- d) Learners to make patterns of their choice using the five shapes.
- e) Learners in groups to make patterns, colour them and share with other groups.
- f) Learners to make patterns using digital devices.

The concept of shapes at this level exposes learners to establishing the rule in a pattern as well as making patterns using shapes. The idea of establishing the rule governing the patterns calls for thinking critically in order to bring out the shapes being repeated. This enhances critical thinking and problem solving skills as well as the motivation to learn more, nurturing the intra-personal skills and values such as self-esteem and confidence.

When learners are given space and time to make patterns using shapes (squares, circles, triangles, ovals and rectangles), different “thought-patterns” would emerge. Consequently these would develop creativity and imaginations as the learner comes up with novel ideas and patterns, making learning more interesting for themselves and the group members.



As they work in groups and create new patterns, learners would appreciate the value of unity among themselves hence enhancing togetherness regardless of their diverse innovations.

As part of non-formal learning, learners could be involved in trimming flower beds to the desired patterns, making patterns in flower gardens as well as identifying and appreciating various patterns on buildings and on clothing in the immediate environment.

The concept of pattern making using shapes is also applicable in other learning areas namely language, environmental, movement and creative activities. The teacher may assess learners through oral questions, written exercises as well as observation using the assessment rubrics stated in the curriculum design.



School Report Card

School

Grade.....Term.....

Name of Learner

Learning Area	CRITERIA <i>(Based on Expected Outcomes in area curriculum design)</i>	Narrative Comments by Learning Area teacher (Based on Rubrics)
Mathematics Activities	• Subtraction of up to two digit numbers • Number patterns	Peter correctly uses subtractions sign (-) to write subtraction sentences. He correctly subtracts using concrete objects and also subtracts up to two digit numbers. He is able to work out missing numbers in number patterns up to 100 correctly.
Environmental Activities		
Language Activities		
Music and movement activities		



CORE COMPETENCES (The teacher to make comments on observations during the term on the learners abilities in the following areas)	
Community Service Learning	
Communication and collaboration Digital literacy	
Critical thinking and problem solving	
Creativity and imagination	
Citizenship	
Any other Comments from other teachers	
Comments by School Head Teacher (To feature Strengths and Targets)	
Comments by Parent/Guardian	

KARANDI PRIMARY SCHOOL

Report of achievement TERM 3 2017 GRADE 2



LEARNER: PETER AMBILA

English Activities:

Peter reads fluently and with excellent comprehension. He is an articulate speaker with a wide oral vocabulary. Peter can ride in a range of contexts with a good awareness of vocabulary, structure and punctuation.

Maths Activities:

Peter has excellent mental and written maths skills. He can select and use appropriate operations in working out solutions to problem. Peter is able to recall multiplication facts reliably.

Environmental Activities:

Peter has a good understanding of properties and reactions. He is able to predict outcomes and record his understanding clearly. Peter enjoys investigations and he is able to explain his thinking with reasoned logic.

Creative Art Activities:

Peter has worked in a range of media, at varying scales. He is developing a detailed observational drawing style.

Music and movement Activities:

Peter is able to play simple tunes on the recorder. He can follow basic musical notation reliably. Peter can identify rhythmic patterns in music and replicate this using percussion.

Physical Education Activities

Peter has developed good team skills in football, Hockey and Rugby. He can sequence movement in dance. James enjoys swimming and able to evaluate performance.

Religious Activities : Peter has a good understanding of the Christian ethos. He can locate books in Bible and is able to recall stories with good details. Peter has learned about the lives of many different saints and he understandings their contributions to our lives.



Strengths and Targets

Peter has a conscientious attitude to his work. He strives to do his best. He is an active learner with advanced thinking skills. Peter should aim to develop the presentation of his work and to continue to use vocabulary and structure to maintain the quality of his writing.

Signed : _____

Class Teacher

Head Teacher

Date